Quadratic Refinements in Normal Play: Realization and the Observation Boundary

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Abstract

Let V be a finite-dimensional vector space over $\mathbb{F}_2$, let B be an alternating form, and let Q be a quadratic refinement of B. We construct an impartial normal-play arena whose loaded position at $x \in V$ is a P-position exactly when $Q(x) = 0$. The board and loading are independent of the refinement; each transition consults at most one value of Q on a fixed coordinate vector. A deterministic public Witt frame reduces the strategic graph to a matching, while weighted source pairs transport the diagonal of Q. A pass-free FIFO strategy available to either seat forces every matched pair to overlap, and an impartial one-move tail compiles the resulting charge to normal play.

The construction has optimal information complexity. Any transcript-stable exact rule must observe directions spanning x. If observations have coordinate weight at most w, at least $\lceil \mathrm{wt}(x)/w \rceil$ are necessary; block compression attains this bound for every $w \geq 1$. For the Gold forms $Q_a(x) = \mathrm{Tr}(x^{1+2^a})$ over finite nimber fields, nim multiplication, Frobenius, and trace supply the complete quadratic datum. The Arf invariant is then the sign of the second-player win bias. The oriented trace cocycle also gives a game-built Gold–Heisenberg extension whose squares recover Q_a and whose commutators recover its polar form.

Two companion results delimit the construction. Every $\mathbb{Z}/4$-valued Brown refinement is decoded by the four outcome classes of one intrinsic partizan selector. Conversely, an ambient-coherent, coefficient-faithful Clifford datum on all short-game values collapses on torsion and, over common coefficient rings, vanishes entirely. Lean checks the complete finite theorem chain: deterministic Witt framing and loading, the weighted-source root, the trace Gold extension and quaternion cell, the intrinsic partizan selector, and the ambient game-exterior obstruction.

1 Introduction

Many classical impartial constructions produce linear losing sets. Coin turning decomposes into nim heaps, error-correcting codes arise as kernels of Grundy maps, and lexicographic codes admit game-theoretic constructions [4, 9]. The present problem is the quadratic analogue: given a quadratic refinement Q of an alternating binary form, construct one uniform impartial rule whose P-set is the quadric $\{x : Q(x) = 0\}$.

Without an access condition the problem is empty, since any finite subset can be made the set of sinks of an ad hoc acyclic game graph. We therefore require the static arena and loading to be independent of the refinement and meter all refinement observations. The construction meets the smallest per-transition budget: one query at one coordinate vector. Its total observation support is also optimal, and the optimum persists when queries of larger coordinate weight are permitted.

The motivating examples are the Gold trace forms

$$Q_a(x) = \mathrm{Tr}_{\mathbb{F}_{2^m}/\mathbb{F}_2}(x^{1+2^a}).$$

On a nimber field, addition is disjunctive sum, multiplication is Conway's nim product, Frobenius is diagonal multiplication, and trace is an iterated sum [7, 13]. Thus both the input space and its quadratic form are assembled from game operations. Gold functions originate in sequence design [10]; their ranks and Walsh spectra are standard objects in finite-field and coding theory [11].

The main theorem is a synthesis of finite-dimensional linear algebra, a matching strategy, and a fixed-tempo normal-play compiler. The stronger isolated-dummy FIFO conjecture for arbitrary graphs is neither assumed nor proved: a Witt frame places every board used here in the matching-plus-isolates class. This distinction is essential both mathematically and for the formal verification boundary recorded in Section 10.

2 Quadratic and game-theoretic preliminaries

2.1 Quadratic refinements

Let V be an m-dimensional $\mathbb{F}_2$-space. A quadratic refinement of an alternating bilinear form B is a function $Q : V \to \mathbb{F}_2$ such that

$$Q(x + y) = Q(x) + Q(y) + B(x, y). \tag{1}$$

For fixed B, the refinements form a torsor under V^*: every other refinement has the form $Q + \ell$ for a unique linear functional ℓ.

If B is nonsingular and $(a_i, b_i)_{i=1}^r$ is a symplectic basis, the Arf invariant is

$$\mathrm{Arf}(Q) = \sum_{i=1}^{r} Q(a_i)Q(b_i) \in \mathbb{F}_2.$$

It classifies nonsingular binary quadratic forms at fixed dimension [3, 8]. Their zero count is

$$\#\{x : Q(x) = 0\} = 2^{2r-1} + (-1)^{\mathrm{Arf}(Q)} 2^{r-1}. \tag{2}$$

For a degenerate form of polar rank $2r$ on an m-dimensional space, the values are balanced when Q is nonzero on $\operatorname{rad} B$. If $Q|_{\operatorname{rad} B} = 0$, the count is

$$2^{m-1} + (-1)^{\mathrm{Arf}(Q_{\mathrm{ns}})} 2^{m-r-1}, \tag{3}$$

where Q_{ns} is the nonsingular quotient.

Fix an ordered basis $e_1, \dots, e_m$. Put

$$P_B(z) = \sum_{j<k} B(e_j, e_k) z_j z_k, \qquad q_j = Q(e_j). \tag{4}$$

Then

$$Q(z) = P_B(z) + \sum_j q_j z_j. \tag{5}$$

The alternating form is public data; the diagonal vector (q_j) is the additional information carried by the refinement.

2.2 Normal-play semantics

A finite impartial normal-play position is a P-position when the previous player wins, equivalently when the player to move has no winning move. The four outcome classes of a finite partizan game are denoted $\mathcal{N}, \mathcal{P}, \mathcal{L}, \mathcal{R}$: respectively the next player, previous player, Left, or Right wins. We use only ordinary Conway equality and normal-play outcomes [7, 4].

3 The access model

Definition 3.1 (Local realization)**.** Fix the ordered coordinate frame of V. A local realization of quadratic refinements has the following data and properties.

1. The alternating form B, the input x, and the coordinate frame are public. The refinement is accessed through queries $Q(z)$.
2. The position set, loading map, terminal set, and declared outcome convention are independent of Q.
3. Each move predicate and transition together make at most c adaptive queries, each at a vector of coordinate weight at most w_0, where c, w_0 are independent of m.
4. For every refinement Q of every admitted B, the loaded root at x has the target outcome if and only if $Q(x) = 0$.

The definition controls local access but intentionally does not define an informal notion of "natural game." Section 5 gives the representation-invariant information lower bound that follows from exactness, and also explains why a generic fork-based liveness test cannot do more.

4 The weighted-source Witt–FIFO arena

4.1 Public adapted coordinates

Use deterministic Gaussian elimination to choose a basis $f_1, \dots, f_m$ depending only on B. It consists of symplectic pairs and radical vectors, and its only nonzero pairings are

$$B(f_{a_h}, f_{b_h}) = 1.$$

Write

$$f_i = \sum_j C_{ji} e_j, \qquad x = \sum_i y_i f_i, \qquad p_i = P_B(f_i).$$

All of C, y, and p_i are public. Expanding the quadratic form in the adapted basis and using the coordinate formula above gives the identity

$$Q(x) = \sum_i y_i p_i + \sum_j x_j q_j + \sum_h y_{a_h} y_{b_h}. \tag{6}$$

Indeed,

$$Q(x) = \sum_i y_i Q(f_i) + \sum_h y_{a_h} y_{b_h}, \qquad Q(f_i) = p_i + \sum_j C_{ji} q_j,$$

and $\sum_i y_i C_{ji} = x_j$.

4.2 The arena

Load one strategic coin F_i for each $y_i = 1$. Join F_{a_h} to F_{b_h} when both are loaded; these edges have weight 1. For each coordinate with $x_j = 1$, also load a source pair S_j^0, S_j^1 whose edge has weight $q_j = Q(e_j)$. The *potential matching* contains every such strategic and source edge, even when its actual weight is zero. Hence the matching is independent of Q.

A core state is $(U, \mathcal{Q}, k, \sigma)$, where U is the untouched set, $\mathcal{Q}$ is a FIFO queue of open coins, $k \in \mathbb{F}_2$ is a one-step ko bit, and $\sigma \in \mathbb{F}_2$ is the accumulated charge. Initially every loaded coin is untouched, the queue is empty, and $k = \sigma = 0$. The position universe contains coherent states for both values of σ, whether or not both are reachable for a given refinement.

There are two impartial moves.

- OPEN(v) removes v from U and appends it to the queue. If its potential mate is already open, the edge weight is added to σ. Opening into an empty queue sets ko; every other opening clears it.
- CLOSE removes the queue front. It is legal when the queue is nonempty and either ko is clear or U is empty. Closing a strategic coin F_i adds p_i to σ; all other closes have zero charge.

There is no pass. Every coin is opened once and later closed once. The quantity

$$2|U| + |\mathcal{Q}| \tag{7}$$

decreases by one on every move, and the exhausted-board exception to ko ensures that the core never stalls before drainage. Thus every core play has exactly two moves per loaded coin.

At a drained state, add one move to an optionless sink exactly when $\sigma = 1$. This tail is impartial and reads only the stored charge, not the refinement oracle.

4.3 The matching strategy

For a queue front f, call an opponent checkpoint *safe* when either ko protects f or its potential mate is not untouched. Equivalently, every legal close has zero degree from f into U in the public matching.

Theorem 4.1 (Public matching strategy)**.** *For a finite matching plus isolates, either designated seat has a deterministic strategy, depending only on the public matching and order, that makes every close have zero public live degree. Consequently the touch intervals of the endpoints of every matched edge overlap.*

Proof. Fix a public order. While U is nonempty, the designated seat never closes. If the queue is empty, it opens the least untouched coin. Otherwise, it opens the untouched mate of the front when that mate exists, and the least untouched coin when it does not. Once U is empty, it closes.

Induct on the displayed clock. Opening into an empty queue sets ko, so the opponent receives a safe checkpoint. If the front's mate is untouched, opening it erases the front's only possible live neighbour. If no mate is untouched, opening any coin preserves that fact. Thus every move of the designated seat hands the opponent a safe checkpoint. At such a checkpoint, a close, if legal, has live degree zero. An opening returns the turn with the same queue front; the designated seat then opens that front's untouched mate if necessary, restoring safety. When U is empty, every close has live degree zero. This proves the claim for either initial seat.

If the endpoints of an edge did not overlap, the first endpoint opened would close while its mate remained untouched, giving live degree one. The strategy excludes this possibility. $\square$

Theorem 4.2 (Quadratic realization)**.** *The weighted-source Witt–FIFO arena is a local impartial realization with $(w_0, c) = (1, 1)$. For every alternating form B, every quadratic refinement Q, and every input x, its loaded root is a P-position if and only if $Q(x) = 0$.*

Proof. Apply Theorem 4.1 to the full public potential matching. The same strategy works for every assignment of source-edge weights. Every strategic edge and every source pair overlaps, so their total opening charge is

$$\sum_h y_{a_h} y_{b_h} + \sum_j x_j q_j.$$

Every active strategic coin closes exactly once, contributing $\sum_i y_i p_i$. By the adapted-coordinate identity, either designated seat therefore forces the drained charge to equal $Q(x)$.

The core has even length, so after drainage it is the first player's turn. If $Q(x) = 1$, the first player uses the public strategy, reaches charge one, and takes the tail move. If $Q(x) = 0$, the second player uses the same strategy and leaves the first player at an optionless drained state. Hence the root is N in the first case and P in the second.

All static data use only B, x, and the fixed frame. A source opening makes at most the single query $Q(e_j)$ and every other transition is refinement-blind. The arena therefore satisfies Definition 3.1 with $(w_0, c) = (1, 1)$. $\square$

The theorem is a strategy statement. It does not claim that every off-policy play drains with charge $Q(x)$. It also does not use a theorem for arbitrary graphs: deterministic Witt reduction produces exactly the matching class of Theorem 4.1.

5 The observation boundary

We now measure the complete set of refinement observations that determines a rooted outcome, allowing the set to be chosen adaptively.

Definition 5.1 (Transcript stability)**.** Let $S(Q, x) \subset V$ be a finite set of observed directions and let $R(Q, x)$ be the rooted result. The observation interface is *transcript-stable* if

$$Q(z) = Q'(z) \text{ for all } z \in S(Q, x) \quad \Longrightarrow \quad R(Q, x) = R(Q', x)$$

for every two refinements Q, Q' of the same polar form.

Theorem 5.2 (Span lower bound)**.** *Suppose the rooted result is transcript-stable and exact at x for the full torsor of refinements of B. Then*

$$x \in \operatorname{span} S(Q, x).$$

If every observed vector has coordinate weight at most w, then

$$|S(Q, x)| \geq \left\lceil \frac{\operatorname{wt}(x)}{w} \right\rceil.$$

In particular, no constant total number of bounded-weight observations gives an exact family in unbounded dimension.

Proof. If $x \notin \operatorname{span} S$, finite-dimensional separation supplies a linear functional ℓ vanishing on S with $\ell(x) = 1$. Then Q and $Q + \ell$ give the same observation transcript and hence the same rooted result, while their target bits at x differ. This contradicts exactness.

For the coordinate bound, vectors of weight at most w cover at most $|S|w$ coordinates. Their span cannot contain x unless their union covers the support of x. Therefore $\operatorname{wt}(x) \leq |S|w$. $\square$

The theorem is the usual annihilator argument for the linear torsor of quadratic refinements, transported to adaptive transcripts; compare the linear-structures framework for Boolean functions in [6].

Corollary 5.3 (Attainment at every width)**.** *Fix $w \geq 1$. There is a local realization whose distinct observations at x are exactly $\lceil \mathrm{wt}(x)/w \rceil$ vectors of weight at most w. Hence the lower bound of Theorem 5.2 is sharp.*

Proof. Partition $\mathrm{supp}(x)$ into disjoint nonempty blocks $Z_1, \ldots, Z_k$ of size at most w, where $k = \lceil \mathrm{wt}(x)/w \rceil$, and let z_i be their indicator vectors. Define $L : \mathbb{F}_2^k \to V$ by $L(e_i') = z_i$, and pull back the quadratic form:

$$Q' = Q \circ L, \qquad B'(u, v) = B(Lu, Lv).$$

Then Q' is a quadratic refinement of B' and $Q'(\mathbf{1}) = Q(z_1 + \cdots + z_k) = Q(x)$. Apply Theorem 4.2 to the induced instance at $\mathbf{1}$, answering its singleton query $Q'(e_i')$ with the original-frame query $Q(z_i)$. The blocks are disjoint, linearly independent, and sum to x, so the resulting observation set has exactly the asserted size and weights. □

The access model cannot, by itself, make an outcome-only definition of "strategically non-evaluative." The following generic obstruction rules out one tempting test.

Proposition 5.4 (Fork padding)**.** *Every finite normal-play tree can be padded, without changing its root outcome, so that every completed play enters an unavoidable two-action fork whose unique winning action depends on a chosen bit. Thus the existence of a reachable, optimal, or unavoidable refinement-sensitive winning fork does not certify non-evaluative behavior.*

Proof. Take a two-option N-position in which exactly one option is P, and exchange the two options when the chosen bit changes. Replace every original terminal P-node by a one-option wrapper whose child is this fork. The wrapper remains P, so backward induction preserves every ancestor outcome, while every original complete play now reaches the bit-sensitive fork. □

6 Gold trace forms

Let $K_m = \mathbb{F}_{2^m}$ and let $\mathrm{Tr}_m : K_m \to \mathbb{F}_2$ be absolute trace. For $a \geq 0$ and $c \in K_m$, define

$$Q_{a,c}(x) = \mathrm{Tr}_m(cx^{1+2^a}), \qquad Q_a = Q_{a,1}. \tag{8}$$

Its polar form is

$$B_{a,c}(x, y) = \mathrm{Tr}_m\big(c(xy^{2^a} + yx^{2^a})\big). \tag{9}$$

On nimbers, all operations in these formulas are game operations. The finite-field implementation is not an algebraically closed backend; all claims here are made over the displayed finite field.

6.1 The all-exponent diagonal source

Choose a basis $e_0, \ldots, e_{m-1}$ of K_m over $\mathbb{F}_2$ with $e_0 = 1$. Nondegeneracy of the trace pairing gives a unique diagonal dual $\lambda_{a,c}^{(m)} \in K_m$ satisfying

$$\mathrm{Tr}_m(\lambda_{a,c}^{(m)} e_i) = \mathrm{Tr}_m(c e_i e_i^{2^a}) \qquad (0 \leq i < m). \tag{10}$$

Theorem 6.1 (Diagonal source criterion)**.** *For every m, a, c as above,*

$$\mathrm{Tr}_m(\lambda_{a,c}^{(m)}) = \mathrm{Tr}_m(c), \qquad \lambda_{a,c}^{(m)} \in \{w^2 + w : w \in K_m\} \iff \mathrm{Tr}_m(c) = 0.$$

In particular, if $m \geq 2$ is even and $c = 1$, there is a $w_a^{(m)} \in K_m$ such that

$$Q_a(e_i) = \mathrm{Tr}_m\big(((w_a^{(m)})^2 + w_a^{(m)})e_i\big) \quad (0 \leq i < m). \tag{11}$$

Proof. Set $e_0 = 1$ in the defining trace-duality identity. This gives $\mathrm{Tr}_m(\lambda_{a,c}^{(m)}) = \mathrm{Tr}_m(c)$. In a finite field of characteristic 2, the image of $w \mapsto w^2 + w$ is exactly the kernel of absolute trace [14, Theorem 2.25]. This proves the equivalence. For $c = 1$ and even m, $\mathrm{Tr}_m(1) = 0$. □

The source can be constructed recursively in a quadratic tower. Write $K_{2M} = K_M(u) = K_M \oplus uK_M$ with conjugation $\sigma(u) = u + 1$. Relative trace satisfies

$$(1 + \sigma)((A + uB)e) = Be, \qquad (1 + \sigma)((A + uB)ue) = (A + B)e. \tag{12}$$

Consequently, if $e_j^{*,(M)}$ is trace-dual to e_j in K_M, then

$$e_{M+j}^{*,(2M)} = e_j^{*,(M)}, \qquad e_j^{*,(2M)} = (1 + u)e_j^{*,(M)}. \tag{13}$$

These identities give a closed recursion for $\lambda_{a,c}^{(2M)}$ from its two diagonal blocks.

Likewise, if $u^2 + u = \delta$ with $\mathrm{Tr}_M(\delta) = 1$ and $\lambda = A + uB$ has trace zero, seek $w = X + uY$. Then

$$w^2 + w = (X^2 + X + \delta Y^2) + u(Y^2 + Y). \tag{14}$$

First solve $Y^2 + Y = B$. The two choices Y and $Y + 1$ toggle the trace of $A + \delta Y^2$, so exactly one makes the second downstairs equation $X^2 + X = A + \delta Y^2$ soluble. Iteration constructs a source throughout the quadratic nimber tower. Theorem 6.1 is not a claim that an arbitrary scaled component has such a source: the exact obstruction is $\mathrm{Tr}_m(c)$.

6.2 Arf as win bias

Corollary 6.2. *Apply Theorem 4.2 to Q_a on K_m. Its P-loadings are exactly the Gold quadric $\{x : Q_a(x) = 0\}$. If the polar rank is $2r$ and Q_a vanishes on its radical, then*

$$\#P - 2^{m-1} = (-1)^{\mathrm{Arf}((Q_a)_{\mathrm{ns}})} 2^{m-r-1}.$$

If Q_a is nonzero on the radical, the two outcomes are balanced.

Proof. The realization theorem identifies P-positions with zeros of Q_a. The nonsingular and radical zero-count formulas above give the stated census. □

For the unscaled Gold form,

$$\mathrm{rad}\, B_a = \mathbb{F}_{2^{\gcd(2a,m)}}, \qquad \mathrm{rank}\, B_a = m - \gcd(2a, m),$$

by trace adjointness and the fixed-field equation $x^{2^{2a}} = x$; this is the standard Gold rank calculation [11]. The radical flag must therefore accompany the Arf invariant whenever the polar form is singular.

7 The Gold–Heisenberg extension

The same trace monomial defines a structural multiplication without a unary quadratic oracle. Put

$$\phi_{a,c}(x,y) = \mathrm{Tr}_m(cxy^{2^a}), \qquad Q_{a,c}(x) = \phi_{a,c}(x,x). \tag{15}$$

Theorem 7.1 (Gold–Heisenberg group)**.** *On $E_{a,c} = \mathbb{F}_2 \times K_m$, define*

$$(s,x)(t,y) = \big(s + t + \phi_{a,c}(x,y), x + y\big). \tag{16}$$

Then $E_{a,c}$ is a group fitting into a central extension

$$1 \longrightarrow \mathbb{F}_2 \longrightarrow E_{a,c} \longrightarrow K_m^+ \longrightarrow 1.$$

For $z = (1,0)$,

$$(s,x)^2 = z^{Q_{a,c}(x)}, \tag{17}$$

$$[(s,x),(t,y)] = z^{B_{a,c}(x,y)}. \tag{18}$$

Its center is

$$Z(E_{a,c}) = \{(s,r) : s \in \mathbb{F}_2, r \in \mathrm{rad}\, B_{a,c}\}.$$

Hence it is extraspecial exactly when $B_{a,c}$ is nondegenerate and nonzero.

Proof. The map $\phi = \phi_{a,c}$ is biadditive, so

$$\phi(x,y) + \phi(x+y,w) = \phi(y,w) + \phi(x,y+w),$$

the normalized cocycle identity. It proves associativity of the displayed product. The identity is $(0,0)$ and $(s,x)^{-1} = (s + Q_{a,c}(x), x)$. Direct substitution gives the square law; swapping two factors changes the central coordinate by $\phi(x,y) + \phi(y,x) = B_{a,c}(x,y)$, proving the commutator law. An element is central precisely when this last value vanishes for every y, which gives the center formula and the extraspecial criterion. □

Every operation in the extension product is game-built on two registered nimber values: disjunctive sum supplies addition, nim product supplies multiplication, diagonal iteration supplies Frobenius, and trace is an iterated sum. The multiplication contains neither a table of basis diagonals nor a $Q_{a,c}$ query; its squares reveal the quadratic form because that is the defining structure of the extension.

For the unscaled form, put $d = \gcd(2a, m)$ and $R = \mathbb{F}_{2^d} = \mathrm{rad}\, B_a$. If m is a power of two and $d < m$, then m/d is even, so absolute trace vanishes on R. Hence

$$Q_a|_R = 0, \qquad \phi_a|_{R\times R} = 0.$$

The subgroup $\{(0,r) : r \in R\}$ is central, and the quotient is a canonical extraspecial extension of K_m/R of order 2^{1+m-d}. This is a per-field, Frobenius-covariant construction. It is not asserted to be coherent under every inclusion of short-game subgroups; Section 9 shows why that stronger condition would force collapse.

8 Brown refinements as partizan outcomes

Let $q : V \to \mathbb{Z}/4$ satisfy

$$q(x + y) = q(x) + q(y) + 2b(x, y), \tag{19}$$

where $b : V \times V \to \mathbb{F}_2$ is symmetric bilinear. Such forms underlie the Brown invariant and the associated fourth-root Gauss sums [5, 18, 16].

Proposition 8.1 (Canonical binary split)**.** *There are unique maps $\ell : V \to \mathbb{F}_2$ and $Q : V \to \mathbb{F}_2$ such that*

$$q = \widehat{\ell} + 2Q,$$

where $\widehat{0} = 0$ and $\widehat{1} = 1$ in $\mathbb{Z}/4$. The map ℓ is linear, and Q is a quadratic refinement with polar form

$$B_Q = b + \ell \otimes \ell.$$

Moreover, if $W(R) = \sum_x (-1)^{R(x)}$, then

$$\sum_{x \in V} i^{q(x)} = \frac{1+i}{2} W(Q) + \frac{1-i}{2} W(Q + \ell). \tag{20}$$

Proof. Reduction modulo two gives the linear map $\ell = q \bmod 2$; the second binary digit defines Q. The carry identity $\widehat{r + s} = \widehat{r} + \widehat{s} + 2rs$ gives $B_Q = b + \ell \otimes \ell$. Finally,

$$i^{\ell} = \frac{1+i}{2} + \frac{1-i}{2}(-1)^{\ell},$$

which yields the Walsh decomposition. This is the two-component decomposition used for generalized Boolean functions; compare [17]. $\square$

For an ordinary binary quadratic form R, let $\mathcal{A}_R(x)$ denote the impartial arena of Theorem 4.2. Thus

$$o(\mathcal{A}_R(x)) = \mathcal{P} \iff R(x) = 0.$$

Theorem 8.2 (Intrinsic Brown selector)**.** *The single partizan game*

$$\mathcal{B}_q(x) = \{\mathcal{A}_{Q+\ell}(x) \mid \mathcal{A}_Q(x)\} \tag{21}$$

has outcome

$q(x)$	0	1	2	3
$o(\mathcal{B}_q(x))$	$\mathcal{N}$	$\mathcal{R}$	$\mathcal{P}$	$\mathcal{L}$.

Hence the fixed decoder $\mathcal{N}, \mathcal{R}, \mathcal{P}, \mathcal{L} \mapsto 0, 1, 2, 3$ recovers $q(x)$.

Proof. The four binary possibilities are

$q(x)$	0	1	2	3
$\ell(x)$	0	1	0	1
$Q(x)$	0	0	1	1
$Q(x) + \ell(x)$	0	1	1	0.

If Left starts, her sole move enters $\mathcal{A}_{Q+\ell}(x)$ with Right to move, so she wins exactly when $Q(x) + \ell(x) = 0$. If Right starts, his sole move enters $\mathcal{A}_Q(x)$ with Left to move, so he wins exactly when $Q(x) = 0$. The two starter bits give the displayed outcomes. $\square$

This is one intrinsic game, not a synchronized product of two arenas: after the root move, play enters exactly one follower. The four-outcome census recovers the correlated Gauss sum in the Walsh decomposition. The selector remains defined when that sum vanishes, although in that case no Brown phase is assigned by Gauss-sum normalization.

9 The ambient-coherent game-exterior obstruction

The additive group of short partizan game values is not a ring under Conway multiplication. Altman and Lipparini ask whether a different game-theoretic product could produce Clifford algebras [2, Problem 5.3(j)]. We give a negative result for a precise value-level interpretation.

Let $\mathcal{S}$ be the additive group of short-game values, R a commutative ring, C an R-algebra, and $\iota : R \hookrightarrow C$ an injective coefficient map. A coefficient-faithful Clifford datum consists of an additive map $j : \mathcal{S} \to C$ and functions Q, B satisfying

$$j(x)^2 = \iota(Q(x)), \qquad j(x)j(y) + j(y)j(x) = \iota(B(x,y)). \tag{22}$$

It is *ambient-coherent* when it is defined on all of $\mathcal{S}$; equivalently, compatible data on a cofinal directed family of finitely generated subgroups glue to such a datum.

Theorem 9.1 (Game-exterior obstruction)**.** *In every ambient-coherent coefficient-faithful datum,*

$$Q(x), B(x,y) \in \bigcap_{k\geq 0} 4^k R \qquad (x, y \in \mathcal{S}).$$

For every torsion game t and every $x \in \mathcal{S}$,

$$Q(t) = 0, \qquad B(t,x) = B(x,t) = 0.$$

Thus the quadratic and polar data factor through the torsion-free quotient. In particular, Q and B vanish identically when $R = \mathbb{Z}$, when R has characteristic 2, or when $R = \mathbb{Z}/4$.

Proof. Moews's structure theorem makes $\mathcal{S}$ two-divisible and gives only power-of-two finite orders [15]; halving is also described explicitly in [12]. For every k, write $x = 2^k u$ and $y = 2^k v$. Additivity of j and the defining Clifford identities give

$$Q(x) = 4^k Q(u), \qquad B(x,y) = 4^k B(u,v),$$

where coefficient faithfulness is used to return from C to R.

Now let $nt = 0$, with n a power of two, and choose y with $ny = t$. Since $nj(t) = j(nt) = 0$,

$$j(t)^2 = j(t)j(ny) = (nj(t))j(y) = 0,$$

so $Q(t) = 0$. For arbitrary x, choose z with $nz = x$. Then

$$j(t)j(x) + j(x)j(t) = (nj(t))j(z) + j(z)(nj(t)) = 0,$$

so $B(t,x) = 0$; symmetry gives the reversed value. Polarization now yields $Q(x+t) = Q(x)$, proving factorization of the quadratic data.

Finally, $\bigcap_k 4^k R = 0$ in each named coefficient ring. □

The coherence hypothesis is substantive. A quadratic table on a selected, root-incomplete subgroup is not constrained by a root that lies outside that subgroup. Likewise the Gold forms use the field operations internal to a finite nimber core and are not asserted to extend through all short-game inclusions. The positive finite-field construction and the negative ambient theorem therefore address different naturality conditions.

10 Formal and executable boundary

The companion Lean development checks the construction end to end:

- `GoldDiagonal.lean`: quadratic-tower trace blocks, trace-dual reconstruction, and the finite-field Artin–Schreier exact sequence;
- `SymplecticBasis.lean` and `WittFrame.lean`: a complete recursive orthogonal decomposition, flattened to an actual deterministic basis with a certified public partial matching;
- `FifoMatching.lean` and `ImpartialRealizer.lean`: the matching safe-front strategy, exact even clock, and impartial tail;
- `PhysicalDeferred.lean` and `GoldArena.lean`: active-coordinate loading, the public and refinement-weighted matchings, literal second-opening and close charges, two ledger conjugacies, and the literal-root P-position theorem;
- `GoldMatchingAlgebra.lean`: the independent adapted-basis expansion and transported diagonal identity;
- `GoldNoEvaluator.lean` and `GoldBlockCompression.lean`: the span lower bound and the induced block instance attaining it;
- `GoldForkPadding.lean`: outcome-preserving unavoidable fork padding;
- `GoldExtraspecial.lean` and `GoldExtraspecialTrace.lean`: the generic cocycle group, the actual field-trace/Frobenius Gold specialization, the cocycle-preserving $(c, 1)$ basis for every trace-one cardinality-four field, and the resulting multiplicative equivalence from that Gold extension to Mathlib's quaternion group; the endpoint is instantiated on Mathlib's canonical GF(4), with trace-one existence discharged;
- `BrownGame.lean` and `BrownSelectorPGame.lean`: the Brown split and one finite partizan game tree whose four recursively derived outcomes give the displayed decoder;
- `GameExterior.lean`: the $\bigcap_k 4^k R$ coefficient theorem, root-free two-primary torsion collapse, and the named vanishing consequences.

Moews's two-divisibility and two-primary torsion theorem for the additive group of short games remains the cited external input to `GameExterior.lean`; Lean proves every subsequent coefficient and vanishing implication from that hypothesis. The phase-aware compiler in `GoldSemantics.lean` is an independent comparison theorem but is not needed for the impartial tail used here. The Lean sources and executable implementation are distributed with Ogdoad [1].

Lean contains both transition systems. The literal root has score zero; OPEN charges an edge exactly when its second endpoint opens and CLOSE charges the displayed public label. Two involutive state encodings move respectively the remaining close labels and the unresolved edge weights into the deferred ledger. Their step-conjugacy theorems transport complete normal-play strategy trees in both directions. A separate ordered-edge calculation proves that the root potential is exactly the strategic cross term plus the singleton source term, while the close potential is exactly the adapted diagonal term. Thus `gold_literal_root_isP_iff` includes the concrete basis, original-frame transport, physical move rules, one public policy across all source weights, and the normal-play tail rather than assuming any composition.

Mathlib's abstract finite fields are not definitionally identified with the Rust finite-nimber backend. The Lean trace specialization proves the field-level theorem, while implementation agreement remains an executable cross-backend check rather than a premise of the mathematics.

The arbitrary-graph isolated-dummy FIFO proposition remains open. It is a strict strengthening of Theorem 4.1, not a missing hypothesis of Theorem 4.2. No finite computation or formal statement of that conjecture is used as a universal proof.

11 Conclusion

Every finite binary quadratic refinement admits a uniform impartial normal-play realization with refinement-blind statics and one singleton query per refinement-sensitive transition. A public Witt frame reduces the polar interaction to a matching, weighted source pairs transport the diagonal, a safe-front FIFO strategy forces the quadratic charge, and a fixed-tempo tail turns that charge into the P/N distinction. Exact transcript-stable realization requires observations spanning the input; the construction and its block compression attain the resulting $\lceil \mathrm{wt}(x)/w \rceil$ bound at every width.

For Gold forms, the entire datum is built from nimber operations. A trace-dual Artin–Schreier source supplies every unscaled diagonal in even degree, the Arf invariant gives the exact zero-set bias, and the oriented trace cocycle gives a structural extension whose squares and commutators recover the quadratic and polar forms. The Brown selector packages a mod-4 refinement into the four intrinsic partizan outcome classes. At the opposite scale, ambient coherence across all short-game values forces coefficient-faithful Clifford data to collapse. These results identify the viable finite-field construction and its precise global boundary without invoking the open arbitrary-graph FIFO conjecture.

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